ASSETS 2026

See What I Mean

A Technology Probe of AI Assistants for Chart Exploration with Low Vision

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Problem Magnification enlarges details but fragments the chart and loses the overview. AI chart assistants mostly serve screen reader users, yet low-vision individuals prefer using their **residual vision**.

RQ1 How do low-vision individuals experience AI-supported chart interaction grounded in their vision?

→ **Sharing their view helped** under both personas: it complemented vision, reduced effort, and kept independence.

RQ2 Should AI support systems for low-vision individuals adapt their response content?

→ **Adapted answers mattered**: 8 of 12 preferred the AI that adapts its answers to low vision, and 9 of 12 followed its directions.

Adapt both sides:
what the AI sees and what it says

DR1 **Anchor** answers on the chart
Pin, highlight, and annotate in the user's view.

DR2 Design for **verifiability**
Show what the AI refers to, so users can check it.

DR3 **Scale** answers to the action
Quick answers for clicks, detail for typed questions.

DR4 Augment vision, **keep agency**
Keep every exchange user-initiated; the user's view leads.

During exploration: ask, follow, point

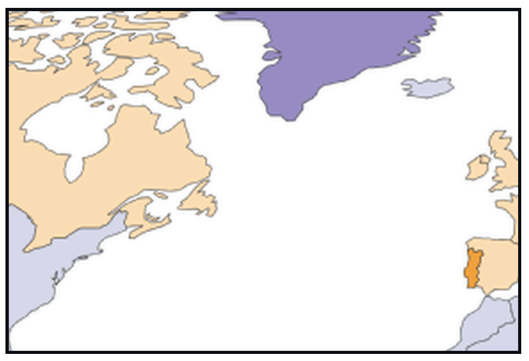
1 Ask



Which countries have the highest female population?

"...Outside your view to the right: Russia, Ukraine, Belarus, and Portugal."

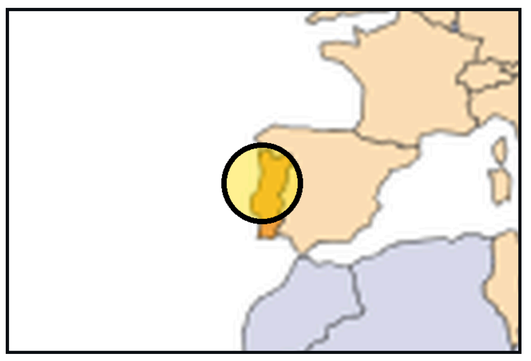
2 Follow guidance



Can you tell me how to find Portugal on this map?

"Portugal is visible on the right edge of your current view, about halfway down."

3 Point to confirm

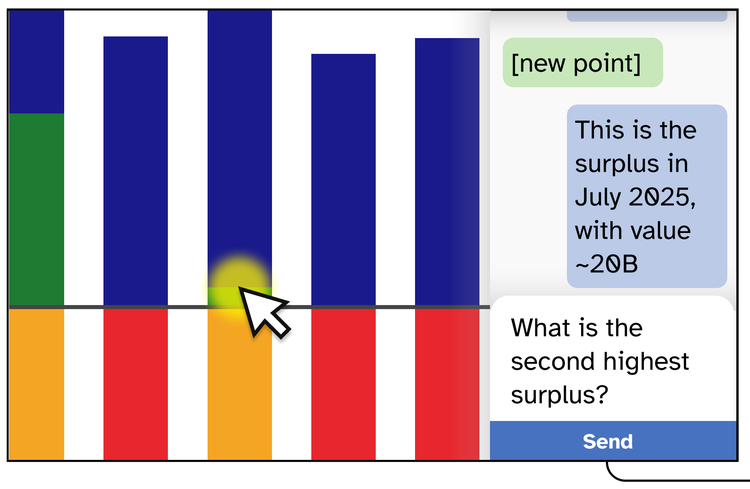


[double-clicks the country]

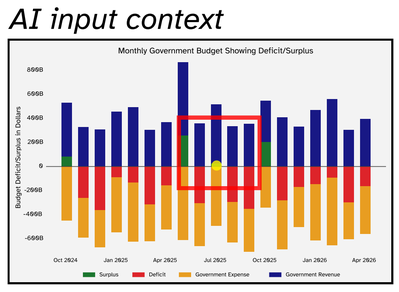
"You are pointing to Portugal. It is colored dark orange..."

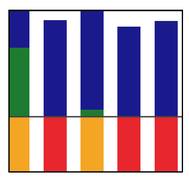
How the probe works: AI shares your view and interest point

Approach The AI assistant sees what you see: your **magnified view** and **point of interest**.



AI input context





Spatial
Semantic
Guidance

General AI (GAI)

The second highest surplus is 205B, in October 2025.

Low-vision AI (LVAI)

The second highest surplus is 205B, in October 2025. It is outside your view, to the right, about one viewport width. The green bar you are pointing is the smallest surplus, 20B, and the highest, 257B, is visible at the left edge of your view.

Technology probe study: 12 low-vision · within-subjects

Training → **Persona A**: chart 1 → **NASA-TLX + custom Likert** →
Persona B: chart 2 → **NASA-TLX + custom Likert** → **SUS + interview**

RQ1 · Seeing my view helped

Complement vision

"...when I asked, it just kind of confirmed that I didn't have to go out of my way to go down and look and piece apart that information. It just kind of laid it out." **P5**

Reduce effort

"usually I have to go scan a chart in, and then manipulate it, and then start looking up the information I'm questioning. This is more of a one-stop shop." **P3**

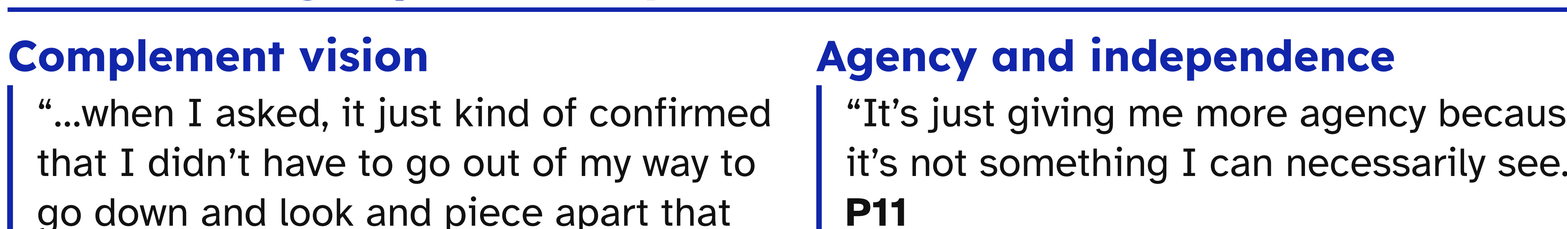
Agency and independence

"It's just giving me more agency because it's not something I can necessarily see." **P11**

"...there's only so many questions you can ask someone before they ask, are you sure you know what you're doing? ... So it's really important to be able to navigate this data independently." **P9**

RQ2 · Adapted answers mattered

Share of answers that include:



"much more specific about tying [the visuals] back to numbers or magnitudes ... better suited to [their] personal needs." **P2**

"...as my vision gets narrower... it's gonna be harder for me to locate [countries]... with this technology, I could click on it, it could tell me, and I could keep moving." **P12**

"describe some of the things on this map that appear very small... what in this chart is potentially visibly difficult to notice." **P11**

Limitation

5 of 12 found the LVAI **wordier**, and it did not align with their in-the-moment needs.

"Sometimes I want an easy answer, value, name." **P1**

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